How type works: anatomy, classification, and voice

Week 3, Lecture 1, Design for Builders: Ship Beautiful Products as a Founder

Autumn 2026

What we will cover

- Why type is the single highest-leverage design decision most builders never make deliberately
- The anatomical terms that explain why two 16px fonts look completely different
- The three main typeface classifications and what each one signals
- How to choose a typeface whose voice matches your brand

Part 1: anatomy

Two fonts, same size, different apparent size

Set Inter and Times New Roman to 16px. Inter looks bigger.

- **X-height** is the height of the lowercase letter x relative to the cap height
- Inter has a large x-height. Times New Roman has a small one
- A large x-height makes text look bigger at the same point size
- It also improves legibility at small sizes and on low-res screens

Practical implication: when you set body size, you are setting x-height, not just pixel count (Segall 2024).

The parts of a letterform

Key terms every product builder needs:

- **Cap height**, height of uppercase letters (H, I, K)
- **Baseline**, the invisible line most letters sit on
- **Descender**, the part of g, y, p that drops below the baseline
- **Ascender**, the part of h, l, k that rises above the x-height
- **Stroke**, any straight or curved line that makes up the letterform

These are not trivia. They determine how much vertical space text actually occupies.

Stress and contrast

Stroke **contrast** = the ratio of thick strokes to thin strokes.

- A high-contrast typeface has thick verticals and thin horizontals (most classical serifs)
- A low-contrast typeface has near-uniform stroke width (most geometric sans-serifs)

Stress = the angle of the thinnest stroke in a rounded letter like O.

- Diagonal stress (axis tilted like /) is typical of humanist and old-style serif typefaces
- Vertical stress (axis straight up) is typical of transitional and modern serifs

High contrast + diagonal stress = historical, editorial voice. Low contrast + vertical stress = technical, neutral voice (Hoffmann 2022).

Part 2: classification

Three classifications that matter for product work

Most typeface classification systems have dozens of categories. For product design, three do the work:

1. **Serif**, strokes end with small cross-lines (serifs)
2. **Sans-serif**, strokes end cleanly, no serifs
3. **Display**, optimized for large sizes, not body text

Within serif and sans-serif, there are important subtypes. Those subtypes carry voice.

Serif subtypes and their voices

Subtype	Example	Voice
Old-style / humanist	Garamond, EB Garamond	Warm, literary, traditional
Transitional	Georgia, Charter	Authoritative, editorial, journalistic
Modern (Didone)	Playfair Display, Didot	Elegant, fashion, high-contrast luxury
Slab serif	Roboto Slab, Courier	Industrial, technical, retro

A fintech product usually reaches for transitional. A fashion brand reaches for modern. An indie blog reaches for old-style.

Sans-serif subtypes and their voices

Subtype	Example	Voice
Grotesque	Helvetica, Inter	Neutral, system-like, professional
Humanist	Gill Sans, Fira Sans	Approachable, warm, readable at body sizes
Geometric	Futura, Nunito, Poppins	Friendly, modern, brand-forward
Neo-grotesque	Arial, Roboto	UI-optimized, digital-native

Most SaaS products use a neo-grotesque or humanist sans for body text. Geometric sans-serifs work well for display and marketing copy but can tire readers at body sizes (Segall 2024).

Display typefaces: use sparingly

Display typefaces are designed for headlines at 40px and above.

- They often have extreme contrast, unusual letterforms, or tight spacing
- They become illegible at body sizes
- One display typeface per product is usually the limit

Do not use a display typeface for body text. The name tells you the use case.

Part 3: voice

What voice means in type

Typography voice is the personality your typeface communicates before a reader processes a single word (Hoffmann 2022).

The reader picks this up in roughly 200 milliseconds. Before they read your headline, they have already felt whether the type is serious, friendly, technical, or cheap.

Choosing a typeface is choosing what your product sounds like.

Matching typeface to brand: a process

Segall (2024) proposes a three-step approach:

1. **Name the brand in adjectives.** Write three words that describe your product's personality. "Precise, modern, trusted" is different from "playful, approachable, helpful."
2. **Match adjectives to classification.** "Precise, modern, trusted" points toward a neo-grotesque or transitional serif. "Playful, approachable, helpful" points toward a geometric or humanist sans.
3. **Test in context.** Set your actual headline copy in the candidate typeface at the actual size. Does it still feel right when it says your words?

Common voice mismatches in builder products

- A comic-app that uses Times New Roman because it was the default (sounds like a legal document)
- A B2B data tool that uses a geometric sans with heavy round letters (sounds like a kids' app)
- A luxury brand using Arial (sounds like a government form)
- A developer tool using a decorative script typeface (sounds confused)

The mismatch is not about aesthetics. It creates cognitive friction: the type says one thing, the product claims another (Hoffmann 2022).

A worked example: choosing a typeface for a productivity app

Product adjectives: “focused, calm, professional.”

Classification match: a neo-grotesque or humanist sans for body text (low contrast, open forms, good legibility at small sizes).

Candidates from Google Fonts: Inter (neo-grotesque), Lato (humanist sans).

Test: set “Capture every idea. Ship every week.” in both at 32px. Inter reads as more system-like. Lato reads as warmer. If the product is a task manager for knowledge workers, Inter. If it is a journaling app, Lato.

Neither is wrong. The choice depends on the adjectives you committed to in step one.

Take-aways

1. X-height determines apparent size. Two typefaces at the same point size look different because of x-height, not pixel size.
2. Stroke contrast and stress axis determine voice. High contrast + diagonal stress = historical. Low contrast + vertical stress = technical.
3. The three working classifications: serif (with subtypes), sans-serif (with subtypes), and display.
4. Each classification carries a voice. Choosing a typeface is choosing what your product sounds like before the reader reads a word.
5. The process: name your brand in adjectives, match adjectives to classification, test in context with real copy.

> “Typography is the spoken language in visual form. Before you read what a piece of type says, you’ve already heard its tone of voice.”
>
> Alma Hoffmann, “Typographic Hierarchies,” Smashing Magazine 2022

What’s next

- **This week, Lecture 2:** Type systems: building a scale, setting line-height and measure, pairing two typefaces
- **This week, section:** Typography audit of your own landing page or product
- **Reading:** Week 3 reading at `/c/design-for-builders-26au/readings/wk03`
- **HW2 assigned this week:** Type-only redesign using Inter, color, and spacing